Date: ________________

Lab # 04 TO UNDERSTAND THE WORKING OF A HALF ADDER AND A FULL ADDER

OBJECTIVES:

O1: To understand working of half adder.

O2: To understand working of full adder.

BACKGROUND THEORY

In electronics, an adder or summer is a digital circuit that performs addition of numbers. In many computers and other kinds of processors, adders are used not only in the arithmetic logic unit(s), but also in other parts of the processor, where they are used to calculate addresses, table indices, and similar operations.

- **Half Adder:**
  The half adder adds two single binary digits A and B. It has two outputs, sum (S) and carry (C). The carry signal represents an overflow into the next digit of a multi-digit addition. The simplest half-adder design, pictured on the right, incorporates an XOR gate for S and an AND gate for C. With the addition of an OR gate to combine their carry outputs, two half adders can be combined to make a full adder.

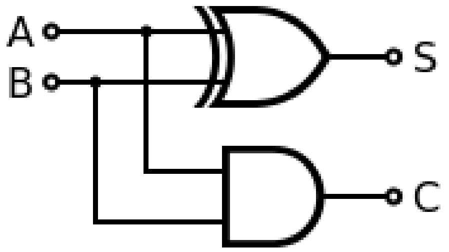

Fig 4.1: Half Adder Circuit

Inputs		Outputs	
Input A	Input B	Sum (S)	Carry (C)
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Table 4.1: Truth Table

Boolean Expressions: $Sum = A \oplus B$, $Carry = A.B$

- **Full Adder:**
  A full adder adds binary numbers and accounts for values carried in as well as out. A one-bit full adder adds three one-bit numbers, often written as A, B, and Carry in (C_{in}); Aand B are the operands, and C_{in} is a bit carried in from the previous less significant stage.The full-adder is usually a component in a cascade of adders, which add 8, 16, 32, etc. bit binary numbers. The circuit produces a two-bit output, output carry and sum typically represented by the signals carry out (C_{out}) and Sum (S).

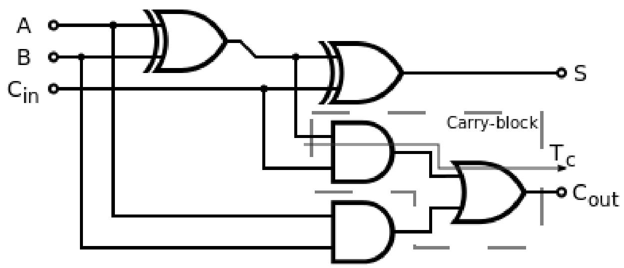

Fig 4.2: Full Adder Circuit

Inputs			Outputs	
Input A	Input B	Carry in (C_{in})	Carry out (C_{out})	Sum (S)
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

Table 4.2: Truth Table

Boolean Expressions: $Sum = A \oplus B \oplus C\,in$

$Carry\ out = (A.B) + (Cin.(A \oplus B))$

HARDWARE REQUIRED:

- Power supply with cables
- Breadboard
- Logic gate ICs: 1) 7486 (XOR gate), 2) NOR Gates, 3) AND Gates,
- LEDs
- Logic Probe (optional)

PROCEDURE:

1. Construct the circuit for Half Adder.
2. Apply all Input combinations to the circuit
3. Check and note the outputs for the corresponding inputs and record the values in the observation table.
4. Now construct the circuit for Full Adder on Breadboard
5. Repeat steps 2-3.

OBSERVATIONS:

Inputs		Outputs	
Input A	Input B	Sum (S)	Carry (C)

Fig 4.3: Truth Table for Half Adder

Inputs			Outputs	
Input A	Input B	Carry in (C_{in})	Carry out (C_{out})	Sum (S)

Table 4.4: Truth Table for Full Adder

Q :) Mention an application both of Half Adder and Full Adder Circuit:

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CONCLUSION (What have you learnt from this experiment?):

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LEARNING OUTCOMES

Upon successful completion of the lab, students will be able to:

LO1: Understand the concept of Half Adder and Full Adder.

LO2: To experimentally verify the behaviour of Half Adder and Full Adder